

Lab 2: Agile Backlog Creation & Sprint Simulation in Jira

PROBLEM STATEMENT #27 | Smart Cities, Transport & Logistics

Warehouse Inventory & Pallet Location Tracker

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Section: D

Epic	Epic Name	Story ID	User Story	Story Pts	Priority
WIPLT-1	Pallet Management	WIPLT-6	Scan Pallet	9	Highest
WIPLT-1	Pallet Management	WIPLT-7	Retrieve Pallet	7	High
WIPLT-2	Storage Validation & Safety	WIPLT-8	Assign Pallet Location	9	Highest
WIPLT-2	Storage Validation & Safety	WIPLT-9	Validate Weight Capacity	7	Highest
WIPLT-3	Inventory Movement Tracking	WIPLT-10	Log Stock Movement	5	Medium
WIPLT-3	Inventory Movement Tracking	WIPLT-11	View Movement History	3	Low
WIPLT-4	Inventory Search & Reporting	WIPLT-12	Search Pallet Location	5	Medium
WIPLT-4	Inventory Search & Reporting	WIPLT-13	Generate Inventory Report	5	Medium
WIPLT-5	Low Stock Monitoring & Alerts	WIPLT-14	Monitor Inventory Levels	5	Medium
WIPLT-5	Low Stock Monitoring & Alerts	WIPLT-15	Send Low Stock Alert	5	Medium

Creating jira workspace

[← Back to templates](#)

Name your space

Required fields are marked with an asterisk *

Name *

Warehouse Inventory & Pallet Location Tracker

How your space is managed *

 Company-managed

▼

[Learn more about space types](#)

☐ Share settings with an existing space

Key * 

WIPLT

Step 1 of 3

Next

Epic 1



For you



Recent



Starred



Apps



Plans



Spaces



Recent



Warehouse Inventory & ...



Recent



WIPLT board

hello

More spaces

Recommended

Collect IT requests

Filters

Search

+ Create

See plans

1

3+

RB

Spaces / Warehouse Inventory ... / WIPLT-1



Pallet Management









Description



Attachments



Add attachment



Create project plan

Try



Child work items



Add child work item



Lichess Local Engine



Details



Assignee



Unassigned



Assign to me



Reporter



Raghav Budur



Labels



None



Due date



None



Start date

Epic 2

 WIPLT |  Epic   

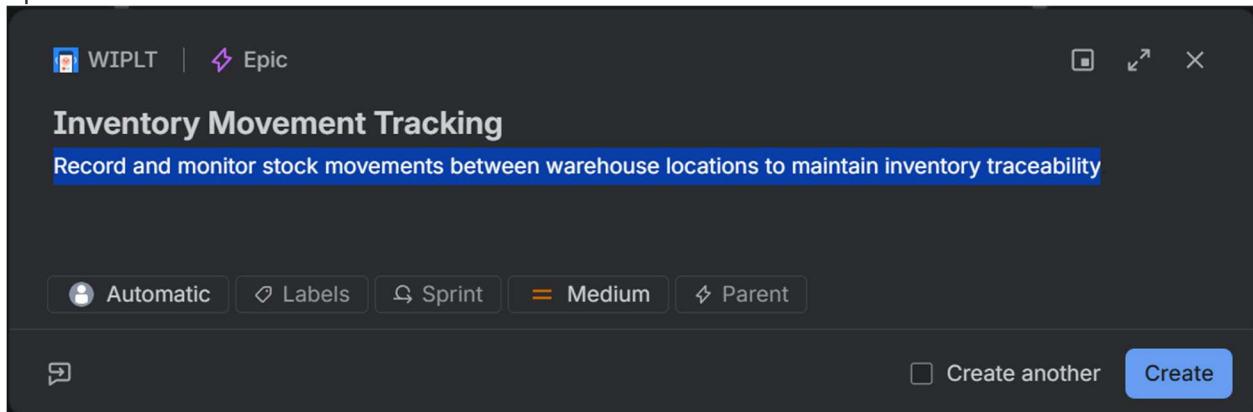
Storage Validation & Safety

Ensure pallets are stored only in locations that comply with rack weight and capacity constraints.

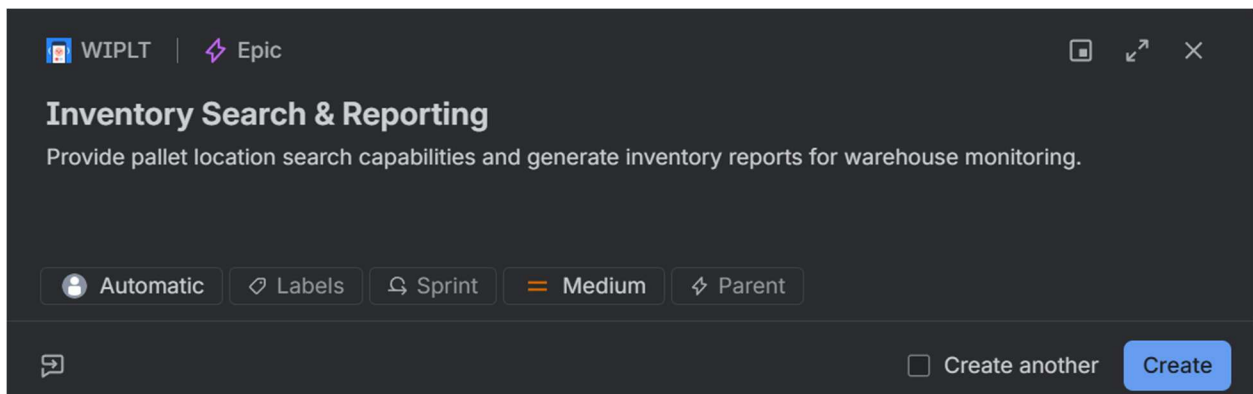
 Automatic  Labels  Sprint  Medium  Parent

 ☐ Create another [Create](#)

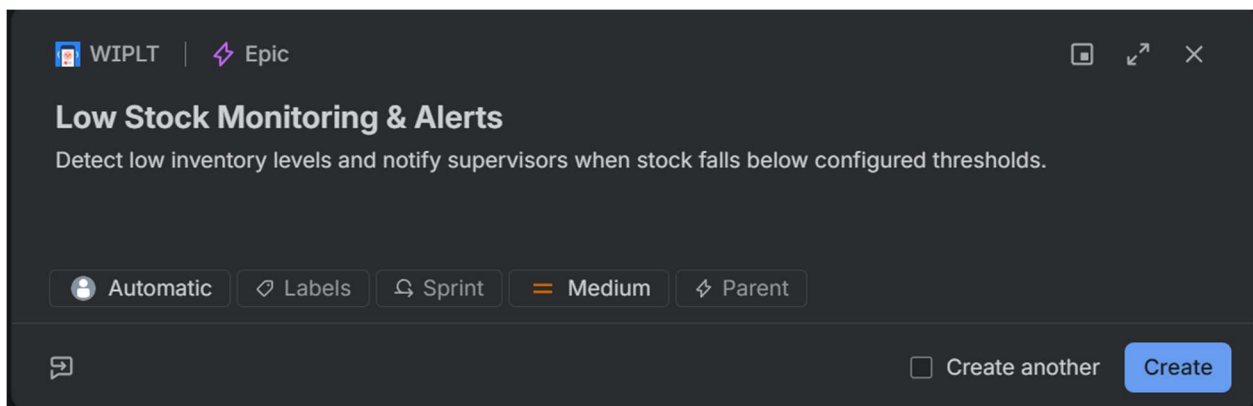
Epic 3



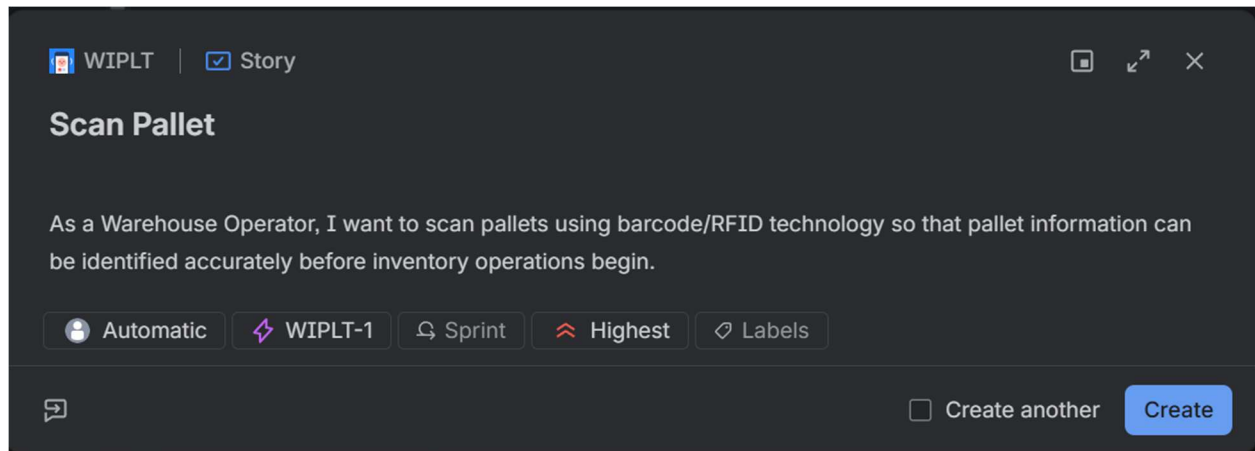
Epic 4



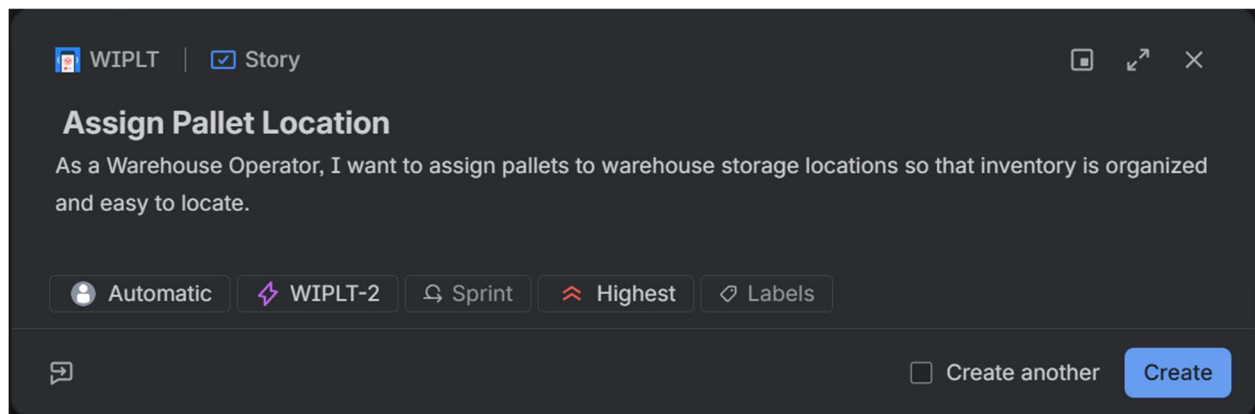
Epic 5



Story 1.1



Story2.1



Story2.2

WIPLT | Story

Validate Weight Capacity

As a Warehouse Operator, I want the system to validate rack weight limits before pallet placement so that unsafe storage conditions are prevented

Automatic

WIPLT-2

Sprint

Highest

Labels

Create another

Create

Story3.1

WIPLT | Story

Log Stock Movement

As a Warehouse Operator, I want stock movements between warehouse locations to be recorded so that inventory traceability is maintained

Automatic

WIPLT-3

Sprint

Medium

Labels

Create another

Create

Story3.2

WIPLT | Story

View Movement History

As a Warehouse Operator, I want to view previous stock movement records so that inventory changes can be tracked and audited efficiently

Automatic

WIPLT-3

Sprint

Low

Labels

Create another

Create

Story4.1

WIPLT | ☒ Story

Search Pallet Location

As a Logistics Supervisor, I want to search for pallet locations so that inventory can be located quickly within the warehouse

Automatic

WIPLT-4

Sprint

Medium

Labels

☐ Create another

Create

Story4.2

WIPLT | ☒ Story

Generate Inventory Report

As a Logistics Supervisor, I want to generate inventory reports so that warehouse stock levels and pallet information can be monitored effectively

Automatic

WIPLT-4

Sprint

Medium

Labels

☐ Create another

Create

Story5.1

WIPLT | ☒ Story

Monitor Inventory Levels

As a Logistics Supervisor, I want inventory levels to be continuously monitored so that low-stock situations can be detected promptly

Automatic

WIPLT-5

Sprint

Medium

Labels

☐ Create another

Create

Story5.2

WIPLT

Story

Send Low Stock Alert

As a Logistics Supervisor, I want the system to send alerts when inventory falls below configured thresholds so that replenishment actions can be initiated in time

Automatic

WIPLT-5

Sprint

Medium

Labels

Create another

Create

All Stories

Backlog (10 work items)

6000

Create sprint

Create a whiteboard to plan your work

Try

WIPLT-6	Scan Pallet	Pallet Management	To Do	9		
WIPLT-7	Retrieve Pallet	Pallet Management	To Do	7		
WIPLT-8	Assign Pallet Location	Storage Validatio...	To Do	9		
WIPLT-9	Validate Weight Capacity	Storage Validatio...	To Do	7		
WIPLT-10	Log Stock Movement	Inventory Move...	To Do	5		
WIPLT-11	View Movement History	Inventory Move...	To Do	3		
WIPLT-12	Search Pallet Location	Inventory Search...	To Do	5		
WIPLT-13	Generate Inventory Report	Inventory Search...	To Do	5		
WIPLT-14	Monitor Inventory Levels	Low Stock Monit...	To Do	5		
WIPLT-15	Send Low Stock Alert	Low Stock Monit...	To Do	5		

Assigning Story points

Backlog (10 work items) 60 0 0 Create sprint

Create a whiteboard to plan your work Try

☒ WIPLT-6 Scan Pallet	Pallet Management	To Do	9	⬆️	👤
☒ WIPLT-7 Retrieve Pallet	Pallet Management	To Do	7	⬆️	👤
☒ WIPLT-8 Assign Pallet Location	Storage Validatio...	To Do	9	⬆️	👤
☒ WIPLT-9 Validate Weight Capacity	Storage Validatio...	To Do	7	⬆️	👤
☒ WIPLT-10 Log Stock Movement	Inventory Move...	To Do	5	=	👤
☒ WIPLT-11 View Movement History	Inventory Move...	To Do	3	⬇️	👤
☒ WIPLT-12 Search Pallet Location	Inventory Search...	To Do	5	=	👤
☒ WIPLT-13 Generate Inventory Report	Inventory Search...	To Do	5	=	👤
☒ WIPLT-14 Monitor Inventory Levels	Low Stock Monit...	To Do	5	=	👤
☒ WIPLT-15 Send Low Stock Alert	Low Stock Monit...	To Do	5	=	👤

Assigning priority

Create a whiteboard to plan your work Try

☒ WIPLT-6 Scan Pallet	Pallet Management	To Do	9	⬆️	👤
☒ WIPLT-8 Assign Pallet Location	Storage Validatio...	To Do	9	⬆️	👤
☒ WIPLT-9 Validate Weight Capacity	Storage Validatio...	To Do	7	⬆️	👤
☒ WIPLT-7 Retrieve Pallet	Pallet Management	To Do	7	⬆️	👤
☒ WIPLT-10 Log Stock Movement	Inventory Move...	To Do	5	=	👤
☒ WIPLT-12 Search Pallet Location	Inventory Search...	To Do	5	=	👤
☒ WIPLT-13 Generate Inventory Report	Inventory Search...	To Do	5	=	👤
☒ WIPLT-14 Monitor Inventory Levels	Low Stock Monit...	To Do	5	=	👤
☒ WIPLT-15 Send Low Stock Alert	Low Stock Monit...	To Do	5	=	👤
☒ WIPLT-11 View Movement History	Inventory Move...	To Do	3	⬇️	👤

Reordering Backlog

Create a whiteboard to plan your work [Try](#)

☒ WIPLT-6	Scan Pallet	Pallet Management	To Do ▾	9	⋈	👤
☒ WIPLT-8	Assign Pallet Location	Storage Validatio...	To Do ▾	9	⋈	👤
☒ WIPLT-9	Validate Weight Capacity	Storage Validatio...	To Do ▾	7	⋈	👤
☒ WIPLT-7	Retrieve Pallet	Pallet Management	To Do ▾	7	⋈	👤
☒ WIPLT-10	Log Stock Movement	Inventory Move...	To Do ▾	5	=	👤
☒ WIPLT-12	Search Pallet Location	Inventory Search...	To Do ▾	5	=	👤
☒ WIPLT-13	Generate Inventory Report	Inventory Search...	To Do ▾	5	=	👤
☒ WIPLT-14	Monitor Inventory Levels	Low Stock Monit...	To Do ▾	5	=	👤
☒ WIPLT-15	Send Low Stock Alert	Low Stock Monit...	To Do ▾	5	=	👤
☒ WIPLT-11	View Movement History	Inventory Move...	To Do ▾	3	▾	👤

Creating sprint:

Sprint 1:

Sprint 1 Setup & Starting Active Board

Edit sprint: WIPLT SPRINT 1

Required fields are marked with an asterisk *

Sprint name *

WIPLT SPRINT 1

Duration

1 week ▾

Start date

9/15/2026

📅

8:15 PM

✕

Date format: MM/DD/YYYY. Time format: e.g. 1:00 PM.

End date

9/22/2026

8:15 PM

Sprint goal

Implement core pallet storage functionality

Cancel

Update

Sprint 1 Work In Progress

WIPLT board Summary Timeline **Active sprints** Backlog Calendar List Forms Development Docs

Search board Version Epic Quick filters Complete sprint Group: Story

TO DO 7	IN PROGRESS 1	DONE 2
Retrieve Pallet Pallet Management WIPLT-7 (7)	Validate Weight Capacity Storage Validation & Safety WIPLT-9 (7)	Scan Pallet Pallet Management WIPLT-6 (9)
Log Stock Movement Inventory Movement Tracking WIPLT-10 (5)	+ Create	Assign Pallet Location Storage Validation & Safety WIPLT-8 (9)
Search Pallet Location Inventory Search & Reporting WIPLT-12 (5)		
Generate Inventory Report Inventory Search & Reporting WIPLT-13 (5)		

Sprint 1 Completion

Complete WIPLT SPRINT 1

This sprint contains **2 completed work items** and **5 open work items**.

- Completed work items includes everything in the last column on the board, **Done**.
- Open work items includes everything from any other column on the board. Move these to a new sprint or the backlog.

Move open work items to

WIPLT SPRINT 2

Cancel Complete sprint

Sprint 2

Edit sprint: WIPLT SPRINT 2

Required fields are marked with an asterisk *

Sprint name *

Duration

Start date

Date format: MM/DD/YYYY. Time format: e.g. 1:00 PM.

End date

Sprint goal

Cancel Update

Starting Sprint 2

Sprint 2 Active Board Progress

Spaces / Warehouse Inventory & Pallet Location Tracker

WIPLT board

Summary Timeline **Active sprints** Backlog Calendar List Forms Development Docs +

Search board Version Epic Quick filters Complete sprint Group: Story

TO DO 5	IN PROGRESS 1	DONE 2
View Movement History Inventory Movement Tracking WIPLT-11 3	Log Stock Movement Inventory Movement Tracking WIPLT-10 5	Validate Weight Capacity Storage Validation & Safety WIPLT-9 7
Search Pallet Location Inventory Search & Reporting WIPLT-12 5	+ Create	Retrieve Pallet Pallet Management WIPLT-7 7
Generate Inventory Report Inventory Search & Reporting WIPLT-13 5		
Monitor Inventory Levels Low Stock Monitoring & Alerts WIPLT-14 5		

Sprint 2 Completed



Complete WIPLT SPRINT 2

This sprint contains **2 completed work items** and **6 open work items**.

- Completed work items includes everything in the last column on the board, **Done**
- Open work items includes everything from any other column on the board. Move these to a new sprint or the backlog.

Move open work items to

WIPLT SPRINT 3

Cancel Complete sprint

Sprint 3:

Start Sprint

×

6 work items will be included in this sprint.

Required fields are marked with an asterisk *

Sprint name *

Duration *

Start date *

×

Date format: MM/DD/YYYY. Time format: e.g. 1:00 PM.

End date *

×

Date format: MM/DD/YYYY. Time format: e.g. 1:00 PM.

Sprint goal

Implement search and reporting.

CancelStart

Sprint 3 Active Scrum Board

Spaces / Warehouse Inventory & Pallet Location Tracker

WIPLT board

Summary Timeline **Active sprints** Backlog Calendar List Forms Development Docs +

Search board Version Epic Quick filters Complete sprint Group: Story

TO DO 3	IN PROGRESS 2	DONE 1
View Movement History Inventory Movement Tracking WIPLT-11 3 0	Log Stock Movement Inventory Movement Tracking WIPLT-10 5 0	Search Pallet Location Inventory Search & Reporting WIPLT-12 5 0
Monitor Inventory Levels Low Stock Monitoring & Alerts WIPLT-14 5 0	Generate Inventory Report Inventory Search & Reporting WIPLT-13 5 0	
Send Low Stock Alert Low Stock Monitoring & Alerts WIPLT-15 5 0		
+ Create		

Sprint 3 Completion



Complete WIPLT SPRINT 3

This sprint contains **1 completed work item** and **5 open work items**.

- Completed work items includes everything in the last column on the board, **Done**
- Open work items includes everything from any other column on the board. Move these to a new sprint or the backlog.

Move open work items to

WIPLT SPRINT 4

Cancel Complete sprint

Sprint 4:

Start Sprint

5 work items will be included in this sprint.

Required fields are marked with an asterisk *

Sprint name *

Duration *

Start date *

Date format: MM/DD/YYYY. Time format: e.g. 1:00 PM.

End date *

Date format: MM/DD/YYYY. Time format: e.g. 1:00 PM.

Sprint goal

Inventory monitoring and alerts.

Cancel

Start

Sprint 4 Active Scrum Board

Spaces / Warehouse Inventory & Pallet Location Tracker

WIPLT board

Summary Timeline **Active sprints** Backlog Calendar List Forms Development Docs +

Search board Version Epic Quick filters Complete sprint Group: Story

TO DO	IN PROGRESS 1	DONE 4
+ Create	Send Low Stock Alert Low Stock Monitoring & Alerts WIPLT-15 5 + Create	Log Stock Movement Inventory Movement Tracking WIPLT-10 5 View Movement History Inventory Movement Tracking WIPLT-11 3 Generate Inventory Report Inventory Search & Reporting WIPLT-13 5 Monitor Inventory Levels Low Stock Monitoring & Alerts WIPLT-14 5

Sprint 4 Completion



Complete WIPLT SPRINT 4

This sprint contains **4 completed work items** and **1 open work item**.

- Completed work items includes everything in the last column on the board, **Done**
- Open work items includes everything from any other column on the board. Move these to a new sprint or the backlog.

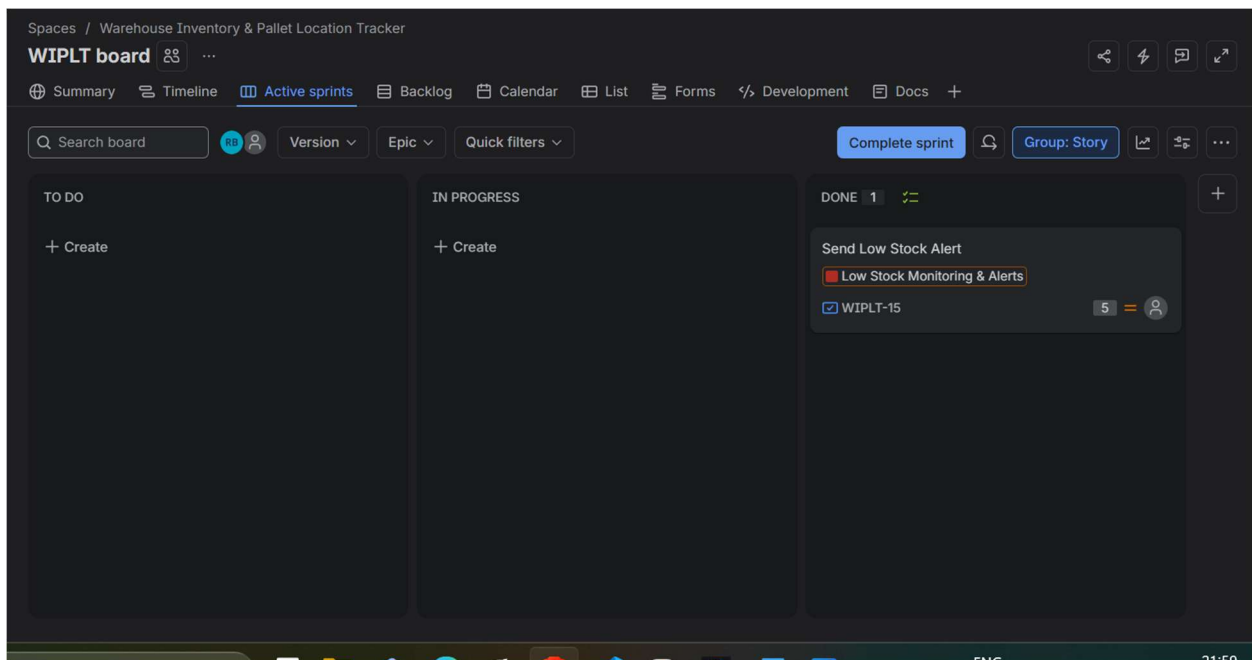
Move open work items to

WIPLT SPRINT 5

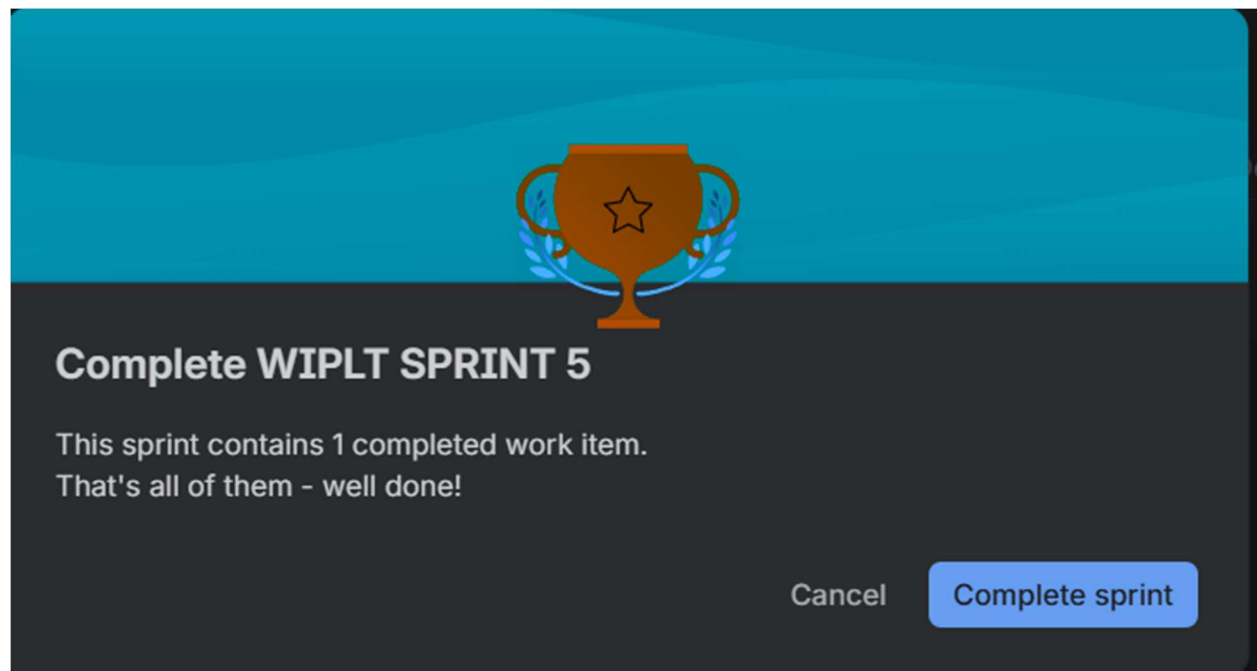
Cancel Complete sprint

Sprint 5:

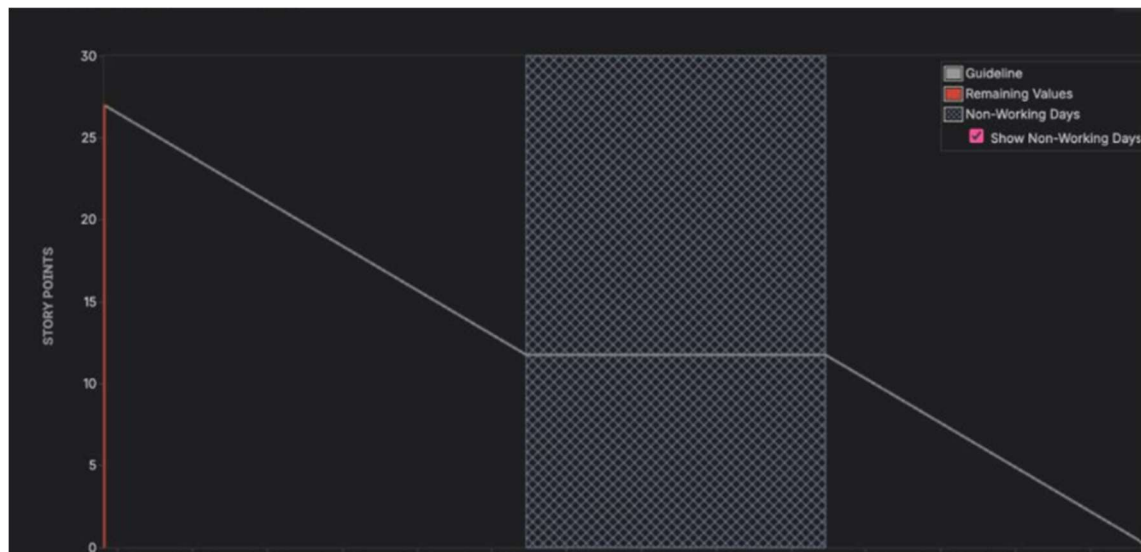
Sprint 5Active Scrum Board

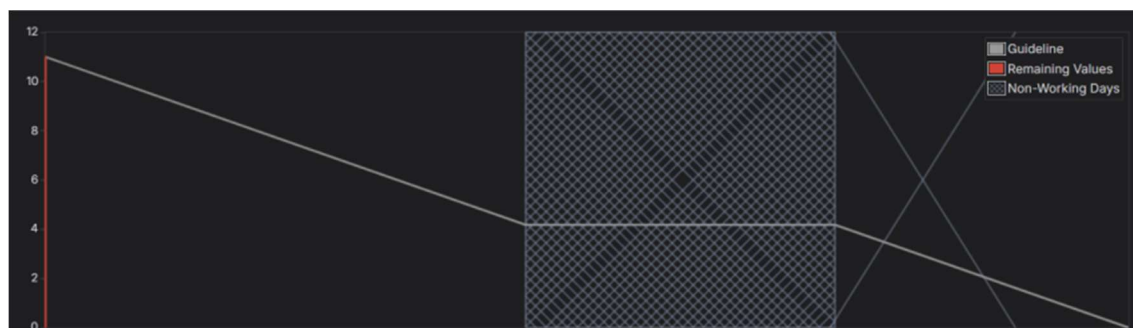
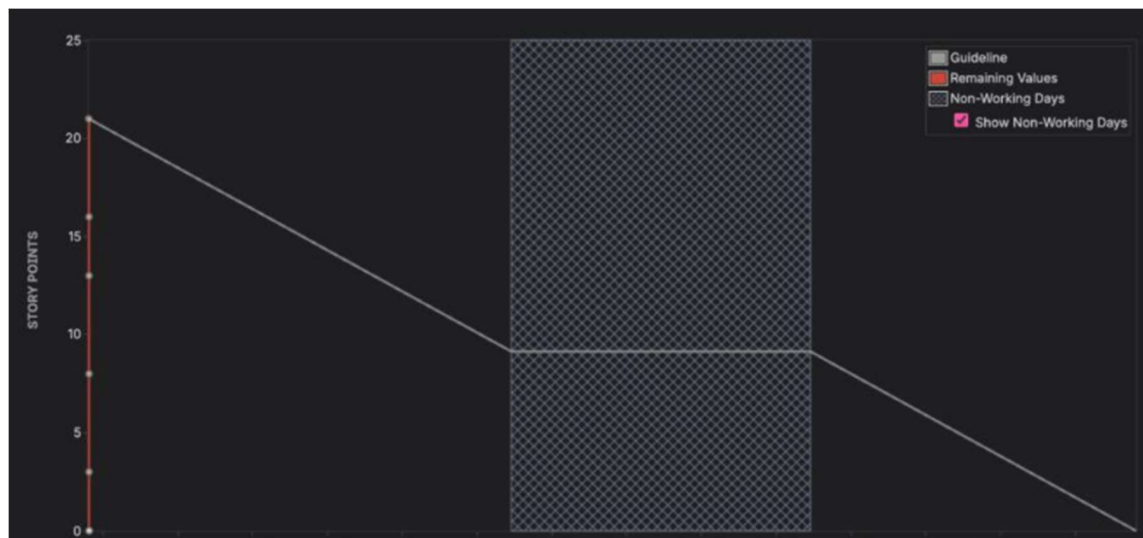


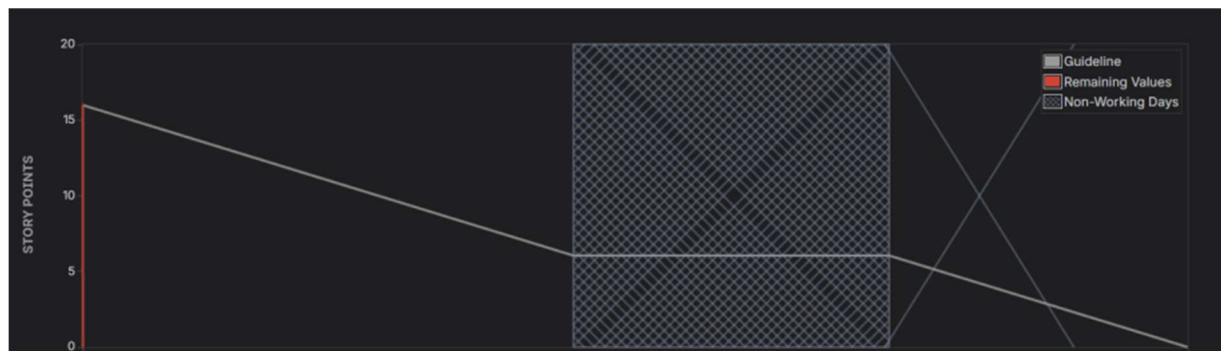
Sprint 5 Completion



BURNOUT







SUMMARY

Sprint report:

SUMMARY:

Work item details

Search table

Work item key	Work type	Work item priority	Issue summary
WIPLT-1		Medium	Pallet Management
WIPLT-10	☒ Story	Medium	Log Stock Movement
WIPLT-11	☒ Story	Low	View Movement History
WIPLT-12	☒ Story	Medium	Search Pallet Location
WIPLT-13	☒ Story	Medium	Generate Inventory Report
WIPLT-14	☒ Story	Medium	Monitor Inventory Levels
WIPLT-15	☒ Story	Medium	Send Low Stock Alert
WIPLT-2		Medium	Storage Validation & Safety
WIPLT-3		Medium	Inventory Movement Tracking
WIPLT-4		Medium	Inventory Search & Reporting

< 1 >

Showing rows 1-15 of 15

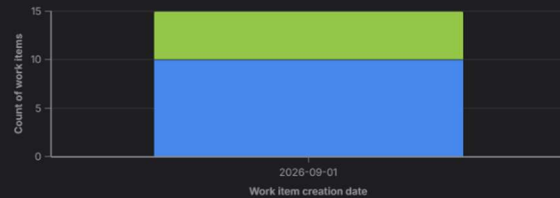
Work item lead time

Story No value

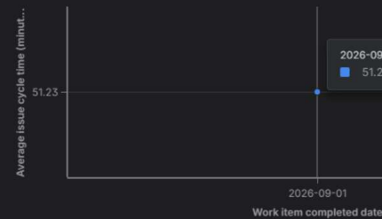


Work item creation trend

Done To Do



Work item cycle time



Group Reflection

This lab gave us a comprehensive and hands-on understanding of how Agile and Scrum methodologies are applied in practice when managing a real-world software engineering project. Before undertaking this activity, concepts such as Epics, User Stories, Fibonacci Story Point Estimation, Sprint Backlogs, and Burndown Charts were primarily theoretical. Working with Jira Software allowed us to directly experience how all these concepts interlock to deliver predictable, high-quality software iterations.

We began by formulating 5 core Epics aligned with the Warehouse Inventory & Pallet Location Tracker problem statement: Pallet Management, Storage Validation & Safety, Inventory Movement Tracking, Inventory Search & Reporting, and Low Stock Monitoring & Alerts. Breaking these down into 10 concise User Stories taught us how high-level architectural requirements are decomposed into user-centric, actionable deliverables with well-defined acceptance criteria.

Estimating each story using Fibonacci story points (3, 5, 8) underscored that estimation reflects technical complexity, risk, and dependencies rather than just calendar time. For instance, Validate Weight Capacity (WIPLT-9) and Monitor Inventory Levels (WIPLT-14) were assigned 8 points due to strict real-time physical safety interlocks and distributed search SLA constraints across 100,000 bins.

Executing the simulated Sprints provided deep insight into the Scrum lifecycle. Moving stories across the Active Board from To Do to In Progress and Done demonstrated how teams visualize workflow, identify bottlenecks, and maintain steady development velocity. We successfully planned and completed 3 balanced sprints, delivering 18, 18, and 16 story points respectively for a total project velocity of 52 story points.

The Burndown Charts were especially valuable for monitoring sprint health in real-time. Comparing the actual burn trajectory against the ideal guideline helped us understand how daily progress is tracked and how early detection of scope creep or stalling tasks ensures that sprint commitments are met on schedule.

Overall, this lab successfully connected theoretical Agile principles with industrial tooling. Using Jira demonstrated that Agile is not just about completing isolated tasks, but about systematic requirement breakdown, collaborative estimation, structured sprint cadence, and transparent progress tracking throughout the software engineering lifecycle.